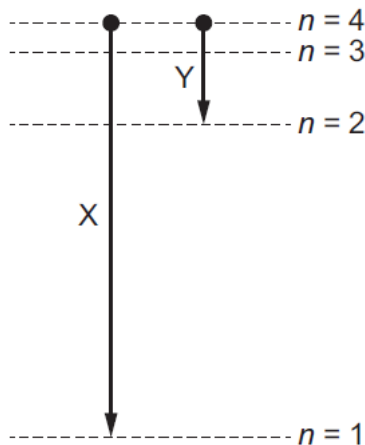


- 28 The diagram shows the lowest four energy levels in the hydrogen atom. The energy levels of the hydrogen atom are given by $E_n = \frac{-13.6 \text{ eV}}{n^2}$.

The electron transitions X and Y result in the emission of two photons.



Which is the ratio $\frac{\text{wavelength of photon due to X}}{\text{wavelength of photon due to Y}}$?

- A $\frac{1}{5}$ B $\frac{1}{4}$ C 4 D 5

Ans: A

$$\Delta E \propto \frac{1}{\lambda} \quad \text{and} \quad \Delta E \propto \left| \frac{1}{n_f^2} - \frac{1}{n_i^2} \right|$$

$$\frac{\lambda_X}{\lambda_Y} = \frac{\Delta E_Y}{\Delta E_X} = \frac{\left(\frac{1}{2^2} - \frac{1}{4^2} \right)}{\left(\frac{1}{1^2} - \frac{1}{4^2} \right)} = \frac{\left(\frac{3}{16} \right)}{\left(\frac{15}{16} \right)} = \frac{1}{5}$$

- 29 In a therapy unit, patients are given treatment from a certain radioactive source. This source has